

The organizational structure of the FAT file system

1. Logical storage of floppy disk data Floppy disks do not need low-level formatting and partitioning operations, just use the FORMAT command for advanced formatting. After formatting, the system will establish the following data structure on the floppy disk: (1) Boot Record (DBR): 0 channels 1 sector on the 0 side, describing the disk structure information. (2) File Allocation Table (FAT): Used to record the allocation of disk space and indicate the information pointer of the cylinder and sector where the hard disk data information is stored. A table entry can be represented in one of four ways: A. A number that represents a pointer to another cluster. B. The number 0 indicates an unused cluster C. A bad sector marker D. File end marker EOF(3) File root table FDT: An index that indicates to deposit data information. Record data such as size, location, date and time of files stored on disk. (4) Data area: store data information.

2. Data organization in the hard disk The new hard drive that has just been purchased from the manufacturer has neither any data nor can it be written, and must be used after low-level formatting, FDISK partition, and advanced FORMAT formatting. This series of initialization work on the hard disk is called hard disk preparation. The process is as follows: Low-level formatting-----FDISK partition-----FORMAT Advanced formatting (1) Low-level formatting: Divide the hard disk into magnetic tracks and sectors, mark the address information on the address field of the sector, and remove the bad magnetic tracks. (2) FDISK: Allows the entire physical hard drive to be logically divided into multiple partitions (up to 4) to realize multiple operating systems to share hard disk space. If the entire physical disk is assigned to DOS/WINDOWS management, the function of the FDISK partition is to divide a physical disk into a primary partition and an extended partition, and then divide the expansion partition into one or more logical disks. While creating a partition table on the hard drive, FDISK writes the master boot record MBR to the main boot record of the hard disk (cylindrical 0, head 0, sector 1) and activates a user-specified partition. (3) FORMAT: Divide the logical sectors in the DOS partition space, generate the DOS boot sector (i.e., logical 0 sector) DBR, file allocation table FAT and root directory table FDT. Under the management of DOS/WINDOWS, the data information is composed of five parts: MBR, DBR, FAT, FDT and data area. Compared with the data structure of floppy disks, hard disks have an additional data area, MBR (Master BootRecord). MBR is called the hard drive master boot record. It is a hard drive boot recording data area established by FDISK on the hard disk cylinder 0, head 0, sector 1, which is used to transfer system control to a certain operating system area specified by the user when the hard disk is booted, and registered in the partition table. MBR DBR FAT1 FAT2 Root Data Area MBR is located at 0 columns, 0 headers, 1 sector DBR starts with logical sector 0 FAT1 starts with logical sector 1

Data organization structure of the hard disk FAT32 file system: MBR
DBR DBR Copy Reserved sector FAT1 FAT2 Root FDT data area MBR
on 0 post, 0 header, 1 sector DBR starts with logical sector 0 DBR
copy starts with logical sector 6 FAT1 starts with logical sector 32

There are two main differences between the data area structure of the hard disk and the floppy disk: (1) There is an MBR independent of the operating system, and (2) The size and start and end sectors of the hard disk are not fixed, which is related to the capacity of the logical disk. While the size of the FAT1, FAT2, and root FDT tables of the hard disk is related to the starting sector number and the capacity of the logical disk. However, the root FDT is a total size of 32 sectors. 2. Hard disk master boot record (DBR) and its structure The 0 cylinder, 0 head, and 1 sector of the hard disk are called the main boot sector, and the content written by the FDISK program to this sector is called the main boot record (MBR). This record occupies 512 bytes and is used to transfer system control to the user when the hard drive boots, and registers an operating system partition in the partition table. The master boot record records the most important structural information about the disk. The primary boot record is created when the hard disk is partitioned (Fdisk). The master boot record includes a small piece of execution code (master boot code), disk characteristics, and a hard disk partition table. The two bytes that end with the main bootstrap record (and the bootstrap sector described later) must be bootstrap markers 0x55AA. The disk feature is located in 0x01B8 and specifies the disk operating system. The floppy disk has no master boot record. The first sector of a floppy disk is the boot sector. Although each hard drive contains a master boot record, the master boot code is used only if it has the active primary partition. The main boot code implements the following functions: * Scan the partition table to find the active partition; * Find the starting sector of the active zone; * Read the boot sector of the active partition to memory; * Run code to execute the boot sector. * If the main boot code does not complete these functions, the system displays the following error message: * Invalid partition table* Error loading operating system* Missing operating system

1. MBR Read: The hard disk boot record (MBR) does not belong to any operating system and cannot be read by the disk operation commands provided by the operating system.

2. The hard disk that constitutes a sector of MBR The MBR consists of 4 parts: (1) the main bootloader (offset address 0000H~0088H), which is responsible for loading and running the system bootloader from the active partition. (2) Error information data area offset address {0089~00E1 is the error message, 10E2H~10BD is 0 bytes) (3) Disk Partition Table (DPT), including 4 partition item offset addresses 01BEH~01FDH, each

The partition table entry is 16 bytes long, and a total of 64 bytes are partition item 1, partition item 2, partition item 3, and partition item 4). Among them, the underlined 16 bytes of data are the parameters of partition item 1 (i.e., the main partition). (4) The end flag offset address (the two byte values of 01EFH~01FFH are the end flag 55AA, if the flag is wrong, the system cannot be started).

3. The partition information structure in the MBR occupies 512 bytes, and the offset address 01BEH~01FDH is 64 bytes, which is the content of 4 partition items (partition information table). It is determined by the disk media type and the user when defining partitions with FDISK. In practical applications, FDISK can divide a hard drive into less than 4 primary partitions, but no less than 4 at most. The item of each partition table is 16 bytes.

4. Main functions and their workflow When starting the PC, the system first tests the hardware equipment, and after success, it enters the bootstrap program INT 19H. Then read the system disk 0 cylindrically, 0 heads, 1 sector of the main boot records the MBR contents to the memory specified unit 0:7C00 first address starting in the area and executes the MBR segment. The boot record of the hard disk does not belong to any operating system, it is transferred to memory and functions before all operating systems, and then gives control to the operating system in the main partition, and lets the main partition information table manage the hard disk. The main functions of MBR segments are as follows: (1) Check whether the hard disk partition table is intact (2) Look for bootable active partitions in the partition table (3) Load the first logical sector contents of the active partition into memory. In DOS/WINDOWS partitions, this sector content is called a DOS boot record. 3. The logic 0 sector of the boot record and its structure disk is called the DOS boot sector, also known as the BOOT zone. We can read out the contents of the DBR sector with the L command in DEBUG. In DEBUG, the L command is formatted as:

-L Memory buffer address, disk number, start sector number, number of sectors to read Enter, where disk A, B, C, D, E... The disk numbers are 0, 1, 2, 3, 4, etc. For example, after entering DEBUG, the command to read the DBR sector of D drive into memory CS:100 is: -L 100, 3, 0, 1 enter

1. The structure of the DBR sector is written to the sector by FORMAT Advanced Formatting to become a DOS boot record DBR, and its main function is to complete the bootstrapping of the DOS/WINDOWS system. Taking the FAT16 hard disk DBR as an example, it consists of the following 5 parts: (1) Jump instruction, which occupies 3 bytes will jump to the boot code, and its content changes with the DOS version. (2) Vendor logo and DOS version number. This section takes up a total of 8 bytes, and its content varies depending on the DOS version. (3) BPB (BIOS Parameter Block, BIOS Parameter Block). BPB takes up 19 bytes from 12th byte. (4) DOS bootstrap. The DOS boot loader is both a 480-byte BOOT code that completes the loading of DOS's three system files. This section varies depending on the DOS version. (5) The end flag word takes up 2 bytes and its value is AA55. The above 5 parts take up a total of 512 bytes and are also exactly one sector, so call it a DOS boot sector or BOOT area. In this section, except for the end mark in Part 5, which is fixed, the other 4 parts are uncertain, parts 1, 2, and 4 are different depending on the DOS version, and the content of the third part will also change with the DOS version and the logic disk parameters of the hard disk.

The BPB parameter block in the DOS boot record occupies 21 bytes from the 12th (0BH) byte, which is an important parameter area of the disk. Reserved Sectors: The number of sectors specified by the operating system that are reserved for boot zones. Media Description: The nature of the disk used provided to the operating system. The value of a media description is a hexadecimal number that DOS can recognize, and different media have different media description values. The media descriptor for the hard drive is "F8". Implicit Sectors: Generally used for hard disk partitions, so in floppy disks this value is 0, in hard disk partitions, it represents the total number of sectors from the disk start sector to the current partition. Each partition of the hard drive is a separate "logical drive", and DOS treats each partition as a continuous sector starting at 0. The implied sectors of each partition depend on the total number of sectors that were previously in that partition. In general, partition 1 has an implied sector count of 17. Main functions of DBR: When the DOS/Windows system boots, the DBR is the first program segment (except for the MBR of the hard disk) to be loaded. Once the DBR is loaded into memory, it begins to execute the boot loader segment, whose main task is to load the DOS system hidden file IO.SYS. The main tasks accomplished by the DBR segment are as follows: (1) reset the boot drive (2) mount the first sector of the root directory (i.e., the first 16 file items in the root FDT) into memory; (3) Check if the first two files in the FDT are DOS two system hidden files. (4) Mount one of the system hidden files IO.SYS into memory. (5) Give control to the system to hide files IO.SYS. Fourth, the document allocation table FAT

File Allocation Table FAT (File Allocation Tabel) is a table used by file management systems to allocate disk physical space to each file, which tells the operating system where files are stored on the disk.

The file allocation phenotypic file system is the most traditional and widely used file system for Microsoft operating systems. The standard FAT uses 16-bit addressing, and the FAT32 scheme was introduced in Windows 95 OSR2. The upper limit of the number of FAT16 clusters is 2 times 16, that is, 65536, and the maximum number of sectors per cluster is 64, so the upper limit of its partition space is 2G (Windows NT/2000 can have 128 sectors per cluster, and FAT16 space can be expanded to 4G); FAT32 clusters use 32-bit addressing, with 4 bits high reserved and the actual number of clusters up to 28 times, and due to the limitations of the current disk physical structure and industry standards, the maximum number of sectors in a disk or disk array is 2 times 32 times, and the space is 2 times 41 times, which is 2TB. Windows 2000 limits the size of FAT32 volumes to 32GB.

The whole purpose of a file system is to keep track of files, and the description needs to describe the usage of each storage unit (cluster) in the entire disk partition, the cluster storage of file data (continuous or fragmented), and the description of the tree directory structure. FAT is actually a mapping table of all cluster usage in a volume, and each file/directory is associated with several items in the table and indexed in the directory. FAT32 is an extension of FAT that supports 32-bit addressing and retains the basic structure of cluster mapping and directory indexing. A File Allocation Table (FAT) consists of a collection of table identifiers and cluster mappings (formerly known as table headings). An identical mirror copy is stored continuously behind the primary FAT table. The table identifier includes the disk media descriptor (1 byte) and the padding byte, which together occupies the cluster mapping area of the two clusters (which is probably why the FAT file system cluster number starts with 2?), and the padding byte, which is usually FFFAT32, also contains "dirty bits", i.e., disk checks (FAT-ChkDsk/NTFS-AutoChk) when the system starts.

FAT is actually made up of FAT12 and FAT16, with cluster mapping values of 12 and 16 bits, respectively, and all other storage logic is the same, which is the standard 16-bit FAT when the total number of clusters is greater than 4080 (FF0h). In the FAT cluster mapping, 0000 represents the empty cluster, FFFF represents the end of the cluster chain, FFF7 represents the bad cluster, and the remaining values represent the cluster number of its subsequent clusters. The storage strategy of FAT12 and FAT32 is similar to that of FAT16, for example, the end cluster of FAT12 is FFF, and the bad cluster of FAT32 is 0FFFFFFF7, etc. (note that FAT32 has a high 4-bit reservation).

FAT has the advantage of being compact and high-speed. With a digitized description of the disk, FAT records the location and region of each file. Naturally, these advantages come at a cost, the price of FAT is the limitation of addressing space (2GB) and the waste of space in the cluster, FAT32's FAT table occupies too much disk space, if the operating system operates a relatively large partition and still tries to put the entire FAT table into memory, it will occupy considerable space and time resources, and will also lead to the fragility of the system, especially when there are many disk fragments, and may even cause the system to crash, at least affecting the robustness of the normal system. Boot sector BIOS parameter table'

The Disk BIOS Parameter Table (BPB) describes the logical structure of the disk, which is the basis for the DOS file system to access files on the disk. The disk BIOS parameter sheet is stored at the beginning of each formatted disk, the disk boot sector, the first sector for floppy disks, and the partition first sector for hard disks, starting at byte 12. The following table is a typical FAT32 boot sector BIOS parameter table (FAT16 includes FAT12):

Offset	Length	Typical Values	Applicable Name	Description
0X0B 2	0002	Bytes per Sector	Size of the hardware sector.	The valid values are 512, 1024, 2048, and 4096, which are generally
5120X0D 1 08	Sectors per cluster	Number of sectors per cluster in the data area.	The operating system can only access a limited number of clusters, and the volume with a large capacity can be accessed by modifying the number of sectors per cluster. The number of sectors per volume cluster depends on the volume capacity, with valid values between 1 and 128 (integer powers of 2. Windows2000 limits FAT32 partitions to a maximum of 32GB, but can access larger partitions established by other operating systems such as Windows95OSR2.	0X0E 2 2000 Reserved Sectors Includes the number of sectors within the boot sector from the beginning of the volume to the first FAT. A typical FAT32 reserved sector is 32 with a FAT of 1.
0X10 1 02	FAT Number	The number of FAT copies of the volume is 2	0X11 2 0000 FAT16 Root Directory Entries The number of 32-byte files and directory name entries stored in the root directory. The hard drive is usually 512, one of which is the volume label. This value is 0	0X13 2 0000 in FAT32 When the number of FAT16 (small) sectors is 16 (<65536). Otherwise it is 0, the number of sectors is defined in the number of sectors in FAT32 this value is 0
0X15 1 1 F8	media descriptor for hard disks is	0XF8 and for high-density 3.5-inch floppy disks is	0XF0. Windows 2000 is no longer used.	0X16 2 0000 FAT16 Number of sectors per FAT Number of sectors per FAT sector, the number of sectors occupied by each FAT in a volume. The operating system uses the number of implied sectors and retains the number of sectors and this parameter. Calculate the root start address and calculate the data area start address based on the number of root entry addresses. FAT32 is 0.
0X18 2 3 F00	Sectors per Track	Number of physical sectors per disk.	INT13H Parameters used to access disks are converted into tracks in multi-head and cylindrical volumes.	0X1A 2 FF00 Number of Heads Number of Heads per Disk. For example, a 3.5-inch 1.44MB floppy disk has 2 heads

OX1C 4 EE39D700 implied Sectors The number of sectors before the volume guides the sector. Used to calculate root eyes during bootstrapping

The absolute address of the record and data area, this value is generally only relevant to INT13H accessible media, and must be 0 in non-partitioned media. OX20 4 7F324E00 (Large) Sector Count If the (small) sector count is 0, the total volume sector is indicated here. If the number of (small) sectors is not 0, this parameter is 0. OX24 8 83130000 FAT32 Number of sectors per FAT The number of sectors occupied by each FAT in a FAT32 volume. The operating system uses the number of implied sectors, retains the number of sectors and this parameter, calculates the root start address, and calculates the data area start address based on the number of root entries OX28 2 0000 FAT32 Extended flag The two-byte structure includes: 0: 3 bits: Number of active FAT (starting from 0, not 1), valid only without mirroring 4: 6 bits: Reserved 7 bits: 0 means FAT is mirrored to all FATs at any time, 1 means only one active FAT (associated with 0..3 bits) 8: 15-bit: Retain OX2A 2 0000 FAT32 file system version High Byte is the main version number, and low byte is the secondary version number. Used to differentiate versions when upgrading to FAT32 in the future, if this value is not 0, lower versions of Windows may wish to questionnaire OX2C 4 02000000 FAT32 root directory cluster number The first cluster number of the root directory, generally 2, but may not be OX30 2 0100 FAT32 file system information sector number The sector number that stores the file system information structure in the FAT32 reserved sector. A typical value is 1. The boot sector's file system information structure is backed up in that sector, but not refreshed OX34 2 0600 FAT32 Boot sector backup A value of non-0 indicates that the boot sector is backed up at the location of the implicit sector, with a typical value of 6, and no other values are recommended OX40 1 80 Physical drive letter Describe the BIOS physical disk number. The floppy disk is 0X00 and the hard disk is 80, which is generally used when the BIOS INT13H calls the disk. This value is valid only for bootable devices. OX41 1 00 Retention This value is 00X42 1 29 Extension Flag The extension boot flag must be 0X28 or 0X29 OX43 4 A88B3652 volume serial number A randomly generated serial number when formatting a disk to help partition the disk OX47 11 NO NAME Label Once used, the label is now stored in the root directory as a special entry OX52 8 FAT32 file system type FAT16 or FAT32 text field

The original disk cardinality table (now abandoned by the operating system) The disk cardinality table stores the basic parameters used to control the hardware when the disk is performing I/O operations, located in the 21h byte of the boot sector. Each item in the table corresponds to one byte. Offset Meaning 21 High 4 bits is the step rate, low 4 bits head unload time 22 Motor wait time (delayed shutdown) 23 Bytes per sector (0-128, 1-256, 2-512, 3-1024, etc.) 24 Number of sectors per channel 25 Number of intersector spacing bytes 26 Number of bytes per sector (when 24 is 0) 27 Fill bytes of the sector interval by formatting operation

28 Formatting operation fills bytes of
sector data area 29 Head stabilization
time after seek 2A Maximum waiting
time for executing a command 2B

1. Size and Position of the FAT Table FAT is arranged on disk immediately after the DOS boot sector DBR, and in FAT16 it always starts with the logical sector 1 of the disk. There are two copies of the FAT table on disk (one is the base FAT table FAT1 and the other is the backup FAT2 of the FAT table), which are lined up closely together on the disk and their size varies depending on the size of the partition. The FAT table is immediately followed by the root directory, and the root directory is followed by the data area. In hard disks, the FAT16 file allocation table is still stored in several sectors starting with logical sector 1, and the number of sectors occupied by a file allocation table varies depending on the hard disk partition and the capacity of the logical disk.

2. After the disk format of the cluster number chain of the file, the user files are stored in the data area in clusters, and a file occupies at least one cluster, when a file occupies multiple clusters, the cluster numbers of these clusters are not necessarily contiguous, but there is an order between these cluster numbers determined by the storage of the file, that is, each file has its own specific cluster number chain. Each available cluster on disk has only one registration entry in FAT, and the cluster in the data area is occupied, idle, or corrupted by populating the table entry value in the registration entry of the corresponding cluster number. Corrupted clusters are discovered during formatting by the FORMAT command, and in a cluster, as long as there is a sector that is problematic, the cluster is unusable. The cluster on disk has 12, 16, or 32 bits of table entries in FAT. In the FAT16 file system, whether the 12-bit or 16-bit table key value taken by each cluster in FAT is related to the capacity of the disk used, the 12-bit item value can represent 4096 clusters, and if the number of clusters on the disk is greater than 4096, the 16-bit item value must be used. In general, floppy disks and DOS partitions of hard drives with less than 20740 sectors map a cluster with 12 bits, such as hard disk partitions with more than 20740 sectors, use 16-bit FAT.

5. File directory table Each formatted disk with files, including 360K, 1.2M and various specifications of hard disks, establishes a file directory table in the disk, which records the useful information of all files on the entire disk, so it is good for us to understand its structure. In the file directory table, each file occupies 32 bytes (two lines when read out with PC-TOOLS or DEBUB), and the meaning of each byte is as follows: 0-7 bytes: file name, a total of 8 bytes, less than 8 are filled with 29, where the first byte is 00, this item is unused, and E5 means this item has been deleted. 8-10 bytes: Extension, 3 bytes, no extension or less than 3 padding with 20. 11 bytes for file attributes. The specific meaning is: 20 is an archive file, 01 is a read-only file, which cannot be modified or deleted. 02 is implicit and cannot be searched with the DIR command, and 04 is a system file, nor can it be searched by the DIR command. 10 indicates that this item is a subdirectory, not a file. 28 indicates that this item is the label of the disk, which is the name of the disk, not a file. 12-21 has a total of 10 bytes as reserved fields and is not used. 22-23 is the time to establish documents. 24-25 is the date of creation of the document.

26-27 is the first cluster number occupied by the file, which is the starting cluster. 28-31 is 4 bytes long for this file.

The position and length of different disk directory tables in the disk are also different, and the 360K disk directory table has a total of 7 sectors from sector 5 to sector 11 relative to the disk. 1.2M disks have a total of 14 sectors in sectors 15 to 28, while 30M drives use DEBUG or PCTOOLS tools software in sectors 123 to 154. For example, use DEBUG to retrieve the 1.2M disk directory table in drive A:

It is easier to read and write with PCTOOLS (5.0). After running, press F3 to enter the disk service state, then press E to enter the edit disk function, select a drive, press F2 and then press R to enter the selection directory area, at this time, press F3 to enter the full-screen editing state after the directory table of the disk is reached, and after the modification is completed, press F5 and then press U to save the disk. Having mastered the structure and modification methods of the directory table, we can use it to do some work, just to name a few. Modify the file name part of a directory item to include spaces and lowercase letters in the middle of the file name. If the file name is modified by this method, the file cannot be called normally under DOS, and if the subdirectory name is modified, the subdirectory cannot be entered normally, because there can be no spaces in the file name and subdirectory name, and the space is used as a separator for commands. This achieves the effect of confidentiality, and when you use it yourself, you only need to change the name to the normal name. If you use this method to modify the volume label item of the hard disk to some lowercase letters, then it can prevent the hard disk from being misformatted, because the volume label name must be entered when the hard disk is formatted, and under normal circumstances, the volume label cannot be entered with lowercase letters. When you really want to format, just change the label to normal. If the file is deleted by mistake and the disk does not perform other write operations, when there is no other software to recover, you can modify the first byte of the directory list and change E5 to other to recover the accidentally deleted file (this method is not the best way to recover files, it is better for ASCII files, and some executable files are prone to errors).

In addition, you can modify the attribute bits, if you modify some files to be implicit or read-only, you can prevent others from deleting and modifying them, and you can prevent virus infection. When encrypting a subdirectory, in order to prevent PCTOOLS from looking up, you can change the length item of the subdirectory to infinity and then change the last item to FF. Other items such as time and date can be modified according to your own needs. Sixth, the zoning table

The partition table determines the basic settings of the disk operating system, including the partition type and location. Each partition table is 16 bytes long and starts at 0x01BE (446) of the master boot record. If there are less than 4 partitions, fill in the remaining 0.

A hard drive can only have a maximum of 4 partitions, and a common logical volume is logically a partition (extended partition), and strictly speaking, a logical volume is not a partition. This is the difference between a volume and a partition. Volumes are a logical concept of boot sector + data area, which is accessed with a colon letter, and a partition may contain one or more volumes. In Windows 2000, only the basic disk uses partition tables. Dynamic disks use the DiskManagement database at the end of the disk to determine disk structure information. The partition table does not change after upgrading to dynamic disks, nor does it change after creating a new dynamic volume.

The partition table can access up to 32-bit sector hard drives. Each sector is 512 bytes (9 bits), so the maximum access to the hard drive is 41 bits (2 TB). The logical volume partition table of an extended partition has up to two records with more columns. Generally speaking, the first partition of the hard disk is the main boot record partition table, the first partition is the basic partition (PRI DOS), and the extended partition (EXT DOS) can be added, and the extended partition contains all the partitions after it, that is, the logical volume.

The first entry of each extended partition table specifies its boot sector, and the second entry specifies the next logical volume. Extended points

Only two entrances are included in the district table, and the third and fourth are not used. A second entry of 0 indicates the end of the logical volume without an extended partition.

The order of each partition in the partition table (including the main partition table and the extended partition table) is irrelevant, and the partition table can also be placed to the last record, and the front is completely empty; If there is an overlap in the space of each partition, the operating system will not judge; The operating system recognizes only the first extended partition in the main partition table and only one logical volume in the logical partition table. Generally, the partition of the hard disk is divided by magnetic tracks, and is not divided across magnetic tracks. Each partition starts from the starting sector of the starting head of a certain magnetic track and ends with the end of the end of the magnetic track. Generally, the number of implied sectors in a partition is the number of sectors on the hard disk; Therefore, the partition starts with a header of 1, the starting sector number is 1, the end header number is the number of hard disk heads minus 1, the end sector number is the number of sectors (the header number starts with 0, the number of sectors starts with 1), and the actual number of starting heads is 0 (i.e., the partition table address).

The starting sector of the first partition in the partition table is the offset relative to the partition table of this partition, which is generally the number of disk sectors, that is, as mentioned above, starting from the second head (No. 1); The starting sector of the second partition is the relative address offset, and for the first partition table (i.e., the 0 head 0 track 1 sector main boot record sector) is the offset of the relative main boot record, which is the same as the absolute address, and the rest of the partition table is the offset of the relative extended partition table, that is, the offset of the second partition starting address. Definition of common system flags in partition tables: System flags description 0x01 FAT12 basic partition or logical drive (less than 32,680 sectors in volume) 0x04 FAT16 partition or logical drive (32,680-65,535 sectors or 16MB-33MB) 0x05 extended partition 0x06 BIGDOS FAT16 partition or logical drive (33MB-4GB) 0x07 installation file system (NTFS partition or logical drive) 0x0B FAT32 partition or logical drive 0x0C FAT32 partition or logical drive using BIOS INT 13h extension 0x0E BIGDOS FAT16 partition or logical drive using BIOS INT 13h expansion 0x0F Expansion partition using BIOS INT 13h expansion 0x12 EISA partition 0x42 dynamic disk volume (when Windows NT 4.0 or earlier creates a mirrored or RAID-5 volume, partition type high position 1) Description of common system flags in partition tables: flags FAT Basic Extension Logic NIT13EXT Supported Capacity Operating System X01 FAT12 yy16M OS2.0 or above win95 and NT X04 FAT16 yy16M-32M OS3.0 or above win95 and NT X05 y2T OS3.3 or above win95 and NT X06 FAT16 yy32M-2G OS4.0 or above win95 and NT X07 NTFS yy2T Windows NT X0b FAT32 yy512M-2T Windows95(OSR2) X0c FAT32 yy y 512M-2T Windows95(OSR2) X0e FAT16 yy y 32M-2G Windows95

X0f and and 2T Windows95

Definition of other system flags: Code 00 01 02 03 04 05 06 07 08 Meaning useless DOS-12 XENIX XENIX DOS-16 EXTEND BIGDOS HPFS SPLIT Code 0A 50 51 53 56 61 63 64 65 75 Meaning BTMGR DM DM DM GB Speed 386/ix NET286 NET386 PCIX Code DB E1 E3 E4 F1 F4 FF The rest? Meaning CM/M Speed Speed Speed Speed Speed Speed bbt MS-DOS can only access partitions with the system flag 0x01, 0x04, 0x05 or 0x06, but can display and delete any type of partition with the Fdisk program. Use the DISK program to access any sector of any partition, such as an NTFS partition.

7. Boot sector The boot sector located at the beginning of each volume is the key disk structure for computer startup. The boot sector contains the code and the data information required to execute the code, including file system information. Boot sectors are generated when the volume is formatted. The last two bytes of the boot sector are the boot markers 0x55AA. The boot sector consists of the following parts: an x86-based CPU jump instruction * Vendor markup (OEM ID) * A data structure, BIOS Parameter Block (BPB) * Extended BIOS Parameter (Extended BPB) * Execution code to boot the operating system * FAT12, FAT16, FAT32 and NTFS boot sectors each contain different formats. * BIOS parameter area describes the physical parameters of the volume; Extended BPB is right next to BPB. FAT12, FAT16, FAT32, and NTFS boot sectors each contain different BIOS parameter zones. * BPB and extended BPB are used to drive disk appliances to read and configure volumes. After extending BPB, when the boot execution* code, the boot execution code boots in the following steps: * System BIOS and CPU power-up self-test* BIOS searches for boot device (such as a disk) * BIOS loads the first physical sector content of the boot device into memory, transmits the memory address to the CUP to interpret the execution If the boot device is a hard disk, the BIOS loads the master boot record, and the main boot code in the master boot record executes the execution code of the active partition boot sector. If there is a floppy disk in drive A, the system BIOS loads the first sector of the floppy disk (boot sector) into memory. If the floppy disk is bootable, the boot sector loads the MS-DOS operating system file IO.SYS into the memory with the execution code for the CPU to execute, and if the floppy disk is unbootable, the execution code displays: Non-System disk or disk errorReplace and press any key when ready. The CMOS setup program of the general computer now allows you to set the boot order. If the hard drive boot is showing similar information, it is possible that the boot sector is corrupted. The boot process is initially independent of the disk format and operating system, and the boot sector executes the code before the file system loads and becomes

Key eigenvalues. Boot sector composition: The main boot record executes the boot sector's CPU instructions, so the first 3 bytes of the boot sector must be executable x86-based CPU instructions, which is a jump instruction that skips the non-executable BIOS Parameter Area (BPB) below. After the jump instruction is an 8-byte OEM ID, flag volume formatting the name and version of the operating system. The OEM ID for Windows95/98/2000 is "MSDOS5.0", and Windows 95 OSR2 and Windows 98 are sometimes "MSWIN4.0" and "MSWIN4.1". OEM ID is followed by BIOS Parameter Area (BPB). The starting position of the BPB is fixed. When the length changes due to changes in BPB format and new content, you can modify the x6 jump instruction accordingly.